Module 1

Unveiling Green Mobility

This module is all about shifting gears towards cleaner, greener transportation. We will dive into the unique challenges and opportunities in a developing economy like India, understand the power of government policies, and discover inspiring solutions for a healthier, more sustainable future.



Unit 1.1: Green Mobility Landscape



Key Learning Outcomes

At the end of this unit, you should be able to:

- Define green mobility and its core principles.
- Identify key global trends and drivers for green mobility.
- Analyse the urgency for transitioning to green mobility.
- Showcase the successful green mobility initiatives.

Green Mobility Landscape

What must you do to shift towards cleaner transportation?

Let us explore the global trends, the powerful drivers, and the urgent need to make this transition a reality.

Buckle up, it is time to take the green road ahead!



Technology innovation:

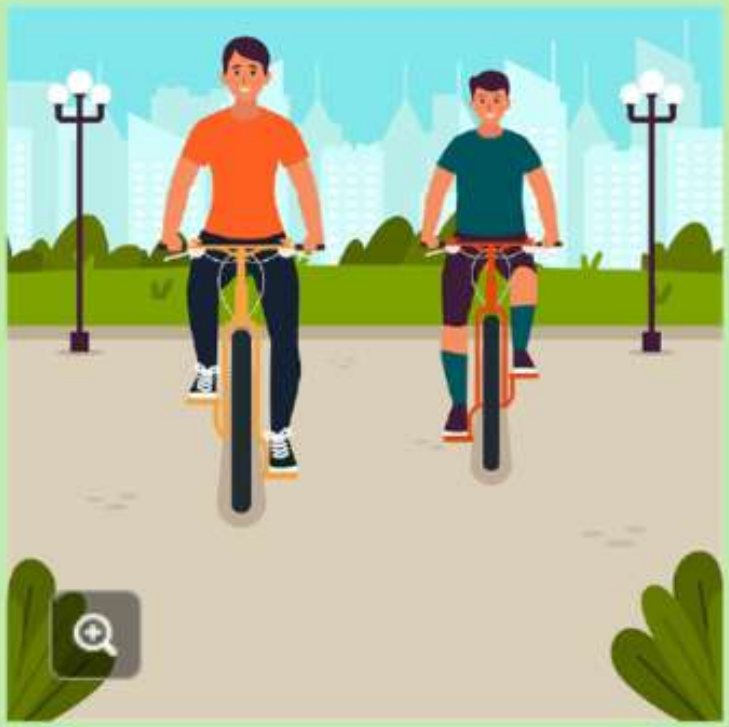
Includes smart technologies like electric vehicles, renewable energy, and AI-powered traffic management for cleaner and smarter transportation.





Land use Planning:

Involves rethinking urban planning around walking lanes, cyclists, and public spaces to create vibrant, liveable cities that breathe free.





Reduced Congestion and Pollution:

Includes reducing congestion through smart infrastructure and alternative modes of transport for smoother, stress-free travel.





Increased Efficiency:

Means optimising public transport, carpooling, micromobility, and walkable cities.





Reduced Reliance on Fossil Fuels:

Means moving away from gasoline and diesel towards renewable energy sources and cleaner alternatives.





1.1.2: Global Trends and Drivers

After understanding green mobility, let us explore the worldwide push behind the shift towards greener transportation solutions.



Trends and Drivers for Green Mobility



Trends and drivers for green mobility include:

1. Rising Environmental Concerns
2. Technological Advancements
3. Policy and Regulatory Framework
4. Economic Opportunities
5. Societal Shift and Changing Consumer Preferences

Let us learn more about each of these.

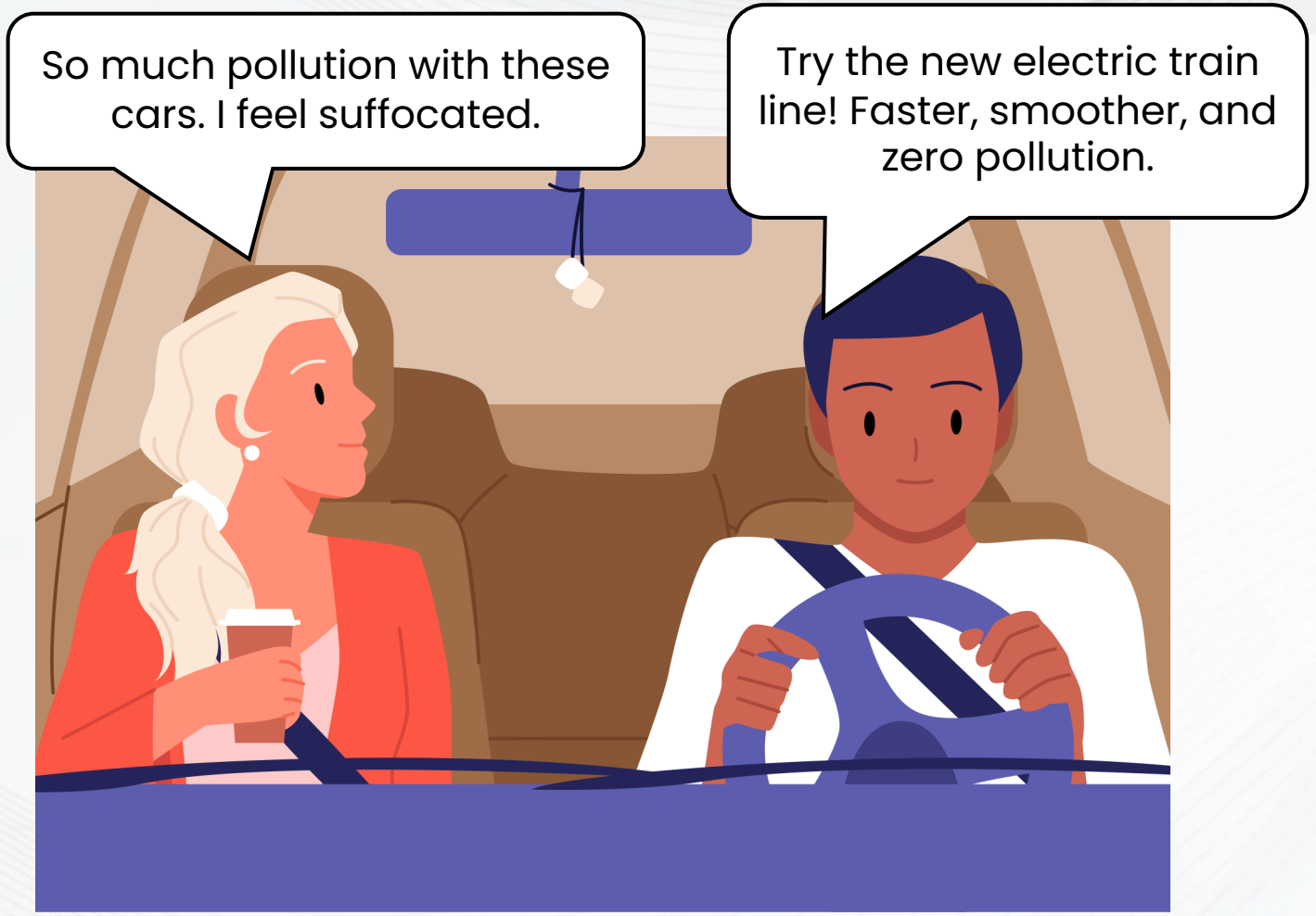


1. Rising Environmental Concerns

Include:

- **Climate Change:** There is growing awareness of the urgency to reduce **greenhouse** gas emissions and address climate change.

- **Air Pollution:** Increasing air pollution levels in major cities, with its harmful health impacts, is pushing for cleaner transportation solutions.



2. Technological Advancements

Include:

Battery Technology: Rapid advancements increasing the **range and performance of electric vehicles**, make them more competitive with gasoline-powered vehicles.

Renewable Energy: Growing availability and affordability of renewable energy sources like solar and wind power is making it feasible to power electric vehicles.



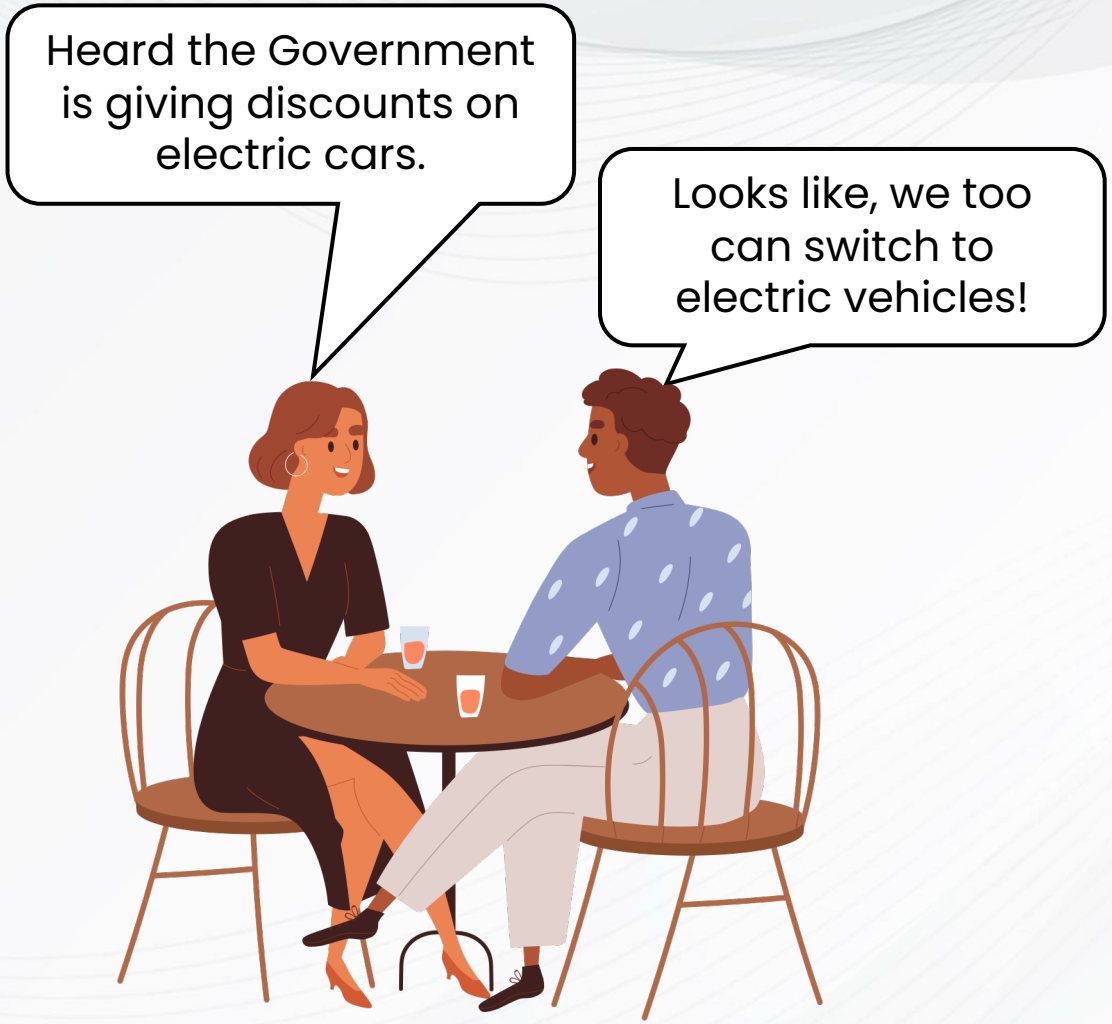
3. Policy and Regulatory Framework

Includes:

- **Government Incentives :** Governments offering subsidies, tax breaks, and incentives to encourage electric vehicle adoption.

- **Stricter Emission Standards:** Pushing manufacturers to develop cleaner technologies.

- **Urban Planning and Infrastructure:** Cities are investing in cycling infrastructure, walkable streets etc.



4. Economic Opportunities

Include:

- **Green Jobs:** New jobs are available in electric vehicle manufacturing, renewable energy, and smart transportation technologies.

- **Energy Security:** Reduces reliance on **fossil fuels** for transportation and reduce dependence on oil-producing nations.

- **Public Health Benefits:** Cleaner air from reduced emissions can improve public health and reduced healthcare costs.



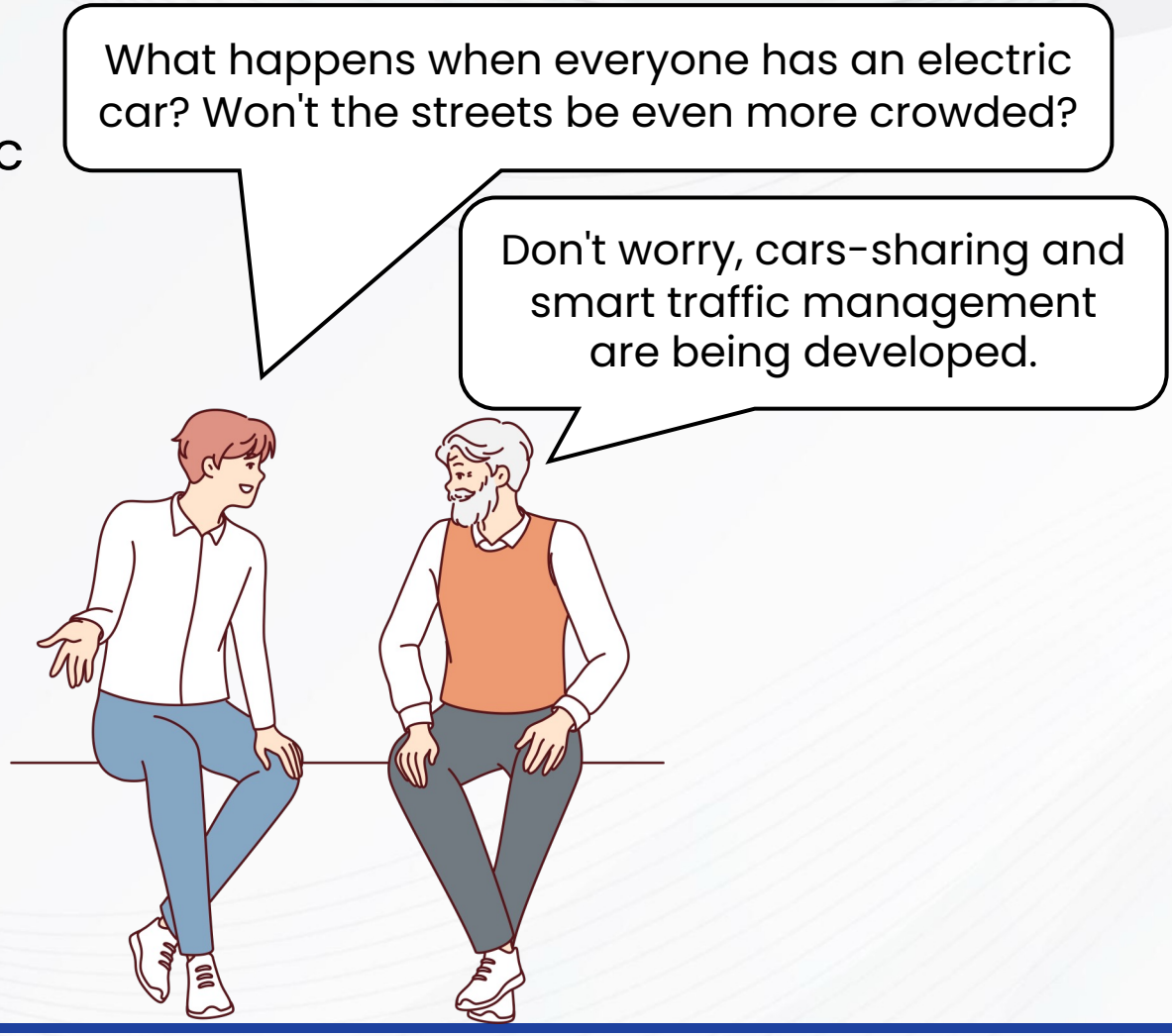
5. Societal Shift and Changing Consumer Preferences

Include:

- **Growing Public Awareness:** Increased public awareness about the environmental and health benefits of green mobility.

- **Shifting Values:** Younger generations are more concerned about **environmental sustainability**.

- **Sharing Economy:** Car-sharing and bike-sharing is providing convenient and affordable alternatives to car ownership.



1.1.3: The Urgency for Green Mobility

The need for green mobility is not just about fancy electric cars and trendy bikes. It is a **dire** necessity fuelled by three urgent challenges to meet. They are:

1. Environmental Crisis
2. Health Risk and Economic Costs
3. Need for Sustainable Solutions

Click next to look at each of them in detail.



Urgency for Green Mobility

Environmental Crisis

Air Pollution

Transportation **generates** air pollution

and

Causes respiratory problems, heart diseases and premature deaths.



Greenhouse Gases

Burning of fossil fuels for vehicles **generates** greenhouse gases

and

Causes global warming, rising sea levels, displacing communities.



Urgency for Green Mobility

Health Risk and Economic Costs

Healthcare Burden

Greenhouse gases result in pollution related illnesses

and

Cause strain in healthcare systems, costing billions in treatment.



Cost of Oil

There is dependency on fossil fuels

and

Makes us vulnerable to price fluctuations and geopolitical instability.



Urgency for Green Mobility

Need for Sustainable Solutions

Urbanisation Boom

Cities are expanding **and** **Cause** worsening of traffic.



Dependence on Oil

Our planet has finite resources **and** **Makes** fossil fuels limited for future use.



Unit 1.2: Green Mobility in India's Context

Key Learning Outcomes

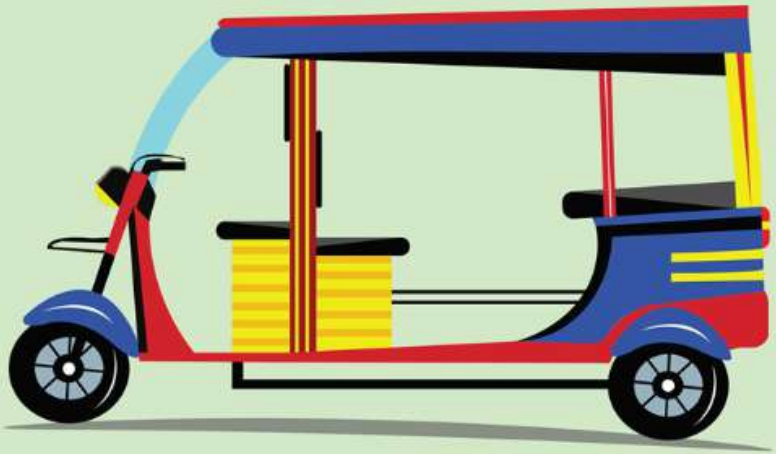


At the end of this unit, you should be able to:

- Describe India's specific urban mobility challenges.
- Evaluate the benefits of implementing green mobility.
- Explore potential economic opportunities with green mobility in India.
- Showcase your understanding of the urban mobility challenges.

Green Mobility in India's Context

Green Mobility is the future of developing countries like India, weaving a web of electric rickshaws, buzzing scooters, and green jobs. Go ahead and discover the potential, and be a part of the ride towards a cleaner, healthier tomorrow!



1.2.1: India's Urban Mobility Challenges

Navigating through India's cities poses daily challenges with crowded streets, honking vehicles, and pollution, impacting the daily lives of millions.



India's Urban Mobility Challenges

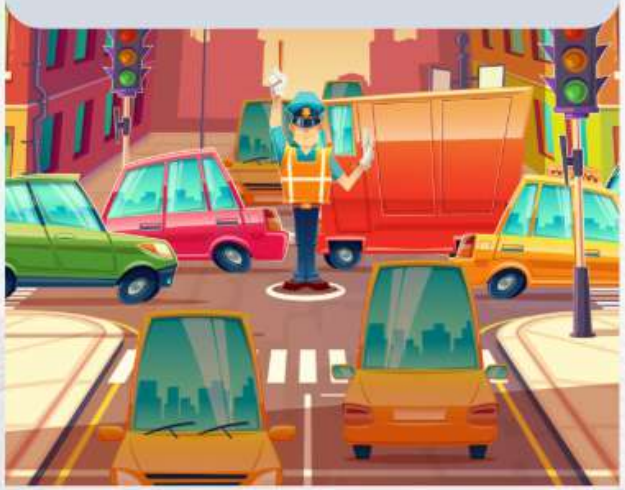
Include:

- Traffic Congestion
- Air Pollution
- Lack of Good Public Transport Infrastructure
- Unsustainable Practices



Traffic Congestion

Jammed streets



Long hours of commuting



Stress on the streets



Air Pollution

Choking on fumes



Health Issues



Lost productivity and healthcare costs



Lack of Public Transport Infrastructure

Inadequate options



Missing connections



Inequality in access



Unsustainable Practices

Fuel dependence



Informal and inefficient systems



Lack of awareness and incentives



1.2.2: Environment and Health Benefits

Green mobility is not just about fancy cars but about transforming our cities into healthier, happier places. Here is how it benefits both the environment and our health.



Environment and health benefits include:

1. Clean Air, Healthy Lives
2. Quieter Streets, Stress-Free Minds
3. Developing Cities, Green Jobs
4. Financial Gains for All
5. A Brighter Future for Generations

Let us look at each of these in detail.



Environment and Health Benefits

Clean Air, Healthy Lives

Imagine:
Delhi's notorious smog replaced with clear skies!

Fact: A study in Bengaluru showed a **20% decrease** in air pollution levels after introducing electric buses, leading to a **15% drop** in hospital admissions for respiratory problems.



Quieter Streets, Stress-Free Minds

Picture:

Bustling streets replaced by the gentle hum of electric vehicles and the chirping of birds.

Fact: A study in Mumbai showed a **50% decrease** in noise levels after introducing electric rickshaws in certain areas, leading to improved sleep patterns and reduced anxiety among residents.



Developing Cities, Green Jobs

Imagine:

Lush parks and green spaces replacing congested parking lots.

Fact: Ahmedabad's BRTS (Bus Rapid Transit System) project created over **10,000 jobs** in construction and maintenance, while reducing traffic congestion and improving air quality.



Environment and Health Benefits

Financial Gains for All

Picture:

Saving money on fuel and parking!

Fact: A study in Chennai showed that switching from a gasoline scooter to an electric scooter could save an average rider **Rs. 1,500** per month on fuel costs as of the year 2023.



A Brighter Future for Generations

Imagine:

Children playing in clean air and enjoying the outdoors without fear of pollution.

Fact: India's ambitious electric vehicle adoption targets aim to reduce carbon emissions by **30% by 2030**, contributing to a cleaner and healthier planet for everyone.



1.2.3: Economic Opportunities

India's green mobility revolution is not just about clean air and quiet streets; it is a **treasure** of economic opportunities waiting to be explored.



PDF **Economic Opportunities**

Economic Opportunities

1. **Job Boom in Green Industries**

Renewable Energy: India's target of 500 GW of renewable energy by 2030 creates a massive demand for skilled professionals.



Electric Vehicle (EV) Manufacturing: India's ambitious EV adoption plans open doors for a thriving domestic manufacturing sector.

Battery Technology: India's focus on developing indigenous battery technology creates opportunities for researchers, engineers, and entrepreneurs.

2. **Infrastructure Development and Upgradation**

Smart cities: Green mobility needs smart infrastructure, creating jobs in intelligent traffic management, charging network development, and data analysis.



Public transport overhaul: Upgrading public transport requires infrastructure development, generating jobs in construction, engineering, and maintenance.

Urban planning and design: Green mobility requires rethinking urban landscapes, and creating jobs in architecture, landscape design, and urban planning.

3. **Sustainable Tourism and Business Opportunities**

Eco-tourism: Green mobility can attract eco-conscious tourists, creating jobs in hospitality, travel agencies, and outdoor activities.



Green startups and entrepreneurship: Green mobility fosters a vibrant startup ecosystem, creating opportunities for entrepreneurs and investors.

Increased efficiency and cost savings: Green mobility can boost productivity and profitability for companies across sectors.

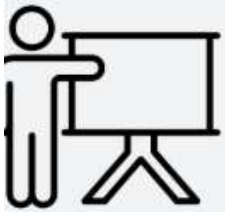
Real-Life Examples:

- Chennai-based **Ather Energy** employs over 1,000 people in its electric scooter manufacturing plant.
- **The FAME-II government scheme** incentivises EV adoption, creating over 5 lakh jobs in the Indian EV sector.
- **Surat's BRTS project** created over 10,000 jobs in construction and maintenance while improving public transport efficiency.

Unit 1.3: Policy Landscape and Governance

Key Learning Outcomes

At the end of this unit, you should be able to:



- Explain government initiatives for promoting green mobility in India.
- Discuss the existing regulatory framework for green mobility in India.
- Identify challenges and opportunities in green mobility implementation.
- Analyse green mobility policies.

Policy Landscape and Governance

Green mobility needs government support.

Now we will explore the policies, regulations, and initiatives shaping India's green mobility journey, and understand how the government is steering us towards a cleaner, greener future.



1.3.1: Government Initiatives and Strategies

Government **initiatives** play a crucial role in shaping the landscape of green mobility in India.

Let us explore some key initiatives and how they impact your daily life:



Government Strategies and Initiatives

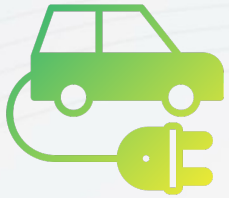


Government Strategies and Initiatives



1. Financial Incentives
2. Public Electrification Push
3. City-Specific Planning
4. Research and Development

Financial Incentives



FAME-II Scheme

This scheme offers subsidies and rebates on electric vehicles (EVs), making them more affordable for individuals and businesses.

State-level Incentives

Many states in India also offer their own incentives for EVs, such as lower registration fees and road tax exemptions.

Public Electrification Push



National Mission on E-Mobility

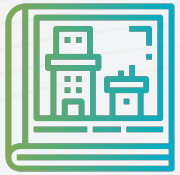
This mission aims to create a robust EV ecosystem in India, including the development of charging infrastructure, battery swapping facilities, and manufacturing of EVs.

State-level Incentives

Several states are investing heavily in charging infrastructure.

For example, Maharashtra aims to install 25,000 charging stations in the next five years.

City-Specific Planning



Smart Cities Mission

This mission focuses on developing smart cities with sustainable transportation systems.

Initiatives like promoting cycling, walking, and public transport, and developing car-free zones.

BRTS Project in Ahmedabad

This project transformed traffic flow in Ahmedabad by creating dedicated bus lanes and walking areas. It has resulted in a significant reduction in travel time and air pollution.

Research and Development



FAME-II	India Energy Storage Alliance
This scheme supports research and development in battery technology, advanced materials, and efficient electric motors.	This alliance seeks to position India as a global leader in battery manufacturing and energy storage solutions.

Reliance Foundation

N·S·D·C RE-IMAGINE FUTURE

Do you know any other strategies and initiatives?



1.3.2: Regulatory Framework and Incentives

The vision of pollution-free vehicles is now becoming a reality, due to a shift in regulatory frameworks and incentives.

Some of the key elements are:

- Emission Norms
- Vehicle Registration and Taxation
- Infrastructure Development Mandates
- Public Procurement Policies
- Research and Development Support



Regulatory Framework and Incentives

Regulatory Framework and Incentives

Some of the key elements are:

1. Emission Norms: Clearing the Air

Tightening the grip	Setting the stage for EVs
The government is tightening emission norms for new vehicle.BS-VI norms, introduced in 2020, are significantly stricter than previous standards.	Stricter emission norms make electric vehicles (EVs), with zero tailpipe emissions, more appealing to manufacturers and consumers.

Example: Delhi has seen a 25% drop in air pollution levels since the introduction of BS-VI norms.



2. Vehicle Registration and Taxation

Faster lanes for EVs	Pushing the market
Several states offer preferential registration and lower taxes for EVs. Delhi exempts EVs from road tax and registration fees.	Encouraging individuals and businesses to choose them over conventional vehicles, boosting their adoption

Example: In Gujarat, an electric car buyer can save up to Rs. 1.5 lakh on registration fees and road tax.

3. Infrastructure Development Mandates

Charging up the city	Making space for sustainability
Building codes and urban planning regulations are being modified to mandate charging infrastructure in new buildings and parking spaces.	Cities are also allocating dedicated parking spots for EVs, providing them with convenient and accessible parking options.

Example: Bengaluru has mandated charging infrastructure in all new commercial buildings.

4. Public Procurement Policies

Government goes green	Setting a trend
Government agencies are prioritising the purchase of EVs for their fleets.	This sends a strong signal to the private sector and citizens, encouraging them to follow suit and adopt EVs.

Example: The Indian government has committed to electrifying 70% of its fleet vehicles by 2030.

5. Research and Development Support

Fueling the future	Unlocking breakthroughs
Government funding initiatives like FAME-II support R&D in battery technology, advanced materials, and efficient electric motors, driving innovation and reducing costs in the EV sector.	These investments are crucial for overcoming technical barriers and making EVs more competitive in terms of performance and price.

Example: The Indian government's support for Research and Development in battery technology has led to the development of affordable lithium-ion batteries.